⚡ JavaScript – Important Interview Q&A (1–30)

1. What is JavaScript?

A high-level, interpreted programming language used to create interactive web applications.

2. Difference between var, let, and const?

- var → function-scoped
- let → block-scoped
- const → block-scoped and cannot be reassigned

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3. What is hoisting?

JavaScript moves variable and function declarations to the top of their scope before execution.

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4. What is a closure?

A function that remembers variables from its outer scope even after the outer function has finished execution.

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5. What is a callback function?

A function passed as an argument to another function.

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6. What is a Promise?

An object representing the result of an asynchronous operation.

States:

- Pending
- Fulfilled
- Rejected

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7. What is async/await?

A cleaner way to handle promises synchronously.

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8. What is the event loop?

Mechanism that handles asynchronous execution in JavaScript.

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9. What is the difference between == and ===?

- == → compares value only
- === → compares value and type

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10. What is the DOM?

Document Object Model — representation of HTML as objects.

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11. What is event bubbling?

Event moves from child to parent.

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12. What is event delegation?

Handling events on a parent element instead of individual children.

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13. What is the spread operator?

... used to copy or expand arrays/objects.

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14. What is destructuring?

Extracting values from arrays/objects into variables.

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15. What are arrow functions?

Short syntax functions without their own this.

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16. What is “this” in JavaScript?

Refers to the object calling the function.

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17. What is prototypal inheritance?

Objects inherit properties from another object.

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18. What is a higher-order function?

A function that takes another function as an argument.

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19. What is map()?

Creates a new array by transforming elements.

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20. What is filter()?

Creates a new array with elements that pass a condition.

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21. What is reduce()?

Reduces array to a single value.

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22. What is debouncing?

Delays function execution until user stops triggering event.

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23. What is throttling?

Limits function execution to once in a specified time.

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24. What is localStorage?

Stores data permanently in browser.

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25. What is sessionStorage?

Stores data for session only.

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26. What is JSON?

JavaScript Object Notation — data format.

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27. What is synchronous vs asynchronous?

- Sync → line by line
- Async → non-blocking

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28. What is try/catch?

Used for error handling.

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29. What is a module in JavaScript?

Reusable file using export and import.

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30. What is ES6?

Modern JavaScript version introducing let, const, arrow functions, classes, modules.

⚛ React – Important Interview Q&A (31–60)

31. What is React?

A JavaScript library for building UI.

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32. What is a component?

Reusable piece of UI.

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33. What is JSX?

JavaScript + HTML syntax.

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34. What are props?

Data passed from parent to child.

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35. What is state?

Internal data that changes inside component.

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36. Difference between state and props?

State is managed inside component; props are passed from parent.

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37. What is useState?

Hook used to manage state.

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38. What is useEffect?

Hook used for side effects like API calls.

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39. What is Virtual DOM?

Lightweight copy of real DOM to improve performance.

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40. What is key in React?

Unique identifier for list items.

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41. What is lifting state up?

Moving state to common parent to share between components.

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42. What is controlled component?

Form element controlled by React state.

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43. What is uncontrolled component?

Form controlled by DOM using refs.

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44. What is React Router?

Library for navigation.

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45. What is Context API?

Global state management without prop drilling.

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46. What is Redux?

External state management library.

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47. What is useRef?

Used to access DOM elements directly.

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48. What is useMemo?

Optimizes performance by memoizing values.

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49. What is useCallback?

Memoizes functions.

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50. What is React Fragment?

Allows grouping elements without extra DOM node.

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51. What is conditional rendering?

Rendering based on condition (if, ternary).

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52. What is React lifecycle?

Mounting → Updating → Unmounting.

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53. What are Hooks?

Functions that let you use state and lifecycle in functional components.

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54. What is prop drilling?

Passing props through many levels.

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55. What is memo?

Prevents unnecessary re-renders.

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56. What is reconciliation?

Process of updating DOM efficiently.

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57. What is Strict Mode?

Helps identify potential issues.

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58. What is lazy loading?

Loads components only when needed.

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59. What is error boundary?

Catches JavaScript errors in components.

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60. What is custom hook?

Reusable hook logic.